

Author Rebuttal

PanAfriCon AI 2026 — Submission 151

Beyond Detection Parity: Localization Parity in Pedestrian Detection

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We thank the Program Committee and both reviewers for their careful reading and constructive feedback. We are encouraged by the accept / weak-accept recommendations and by Reviewer 1's recognition of Localization Parity as a meaningful conceptual contribution. Below we address each comment directly. All requested revisions have been incorporated into the camera-ready manuscript (Springer LNCS / `llncs` format).

Summary of manuscript changes.

- Reformatted the full paper to the official Springer LNCS LaTeX template.
- Named all datasets, detector families, and demographic axes explicitly in the abstract, introduction, and experimental setup.
- Added a formal subsection relating Localization Parity to existing fairness definitions (equalized odds / demographic parity, fair regression, AP-based audits).
- Specified confounders (object scale; occlusion where annotated) and the ICON² + Mann–Whitney statistical protocol.
- Added a discussion of the practical safety implications of 1–3 pp IoU gaps for depth / TTC estimation.
- Clarified that race/skin tone is *not* one of the two audited axes, with justification and future-work plans.

Response to Reviewer 1 (Score: 2 — Accept)

R1.1 — Theoretical relation of Localization Parity to existing fairness definitions

Comment. *Clarify how Localization Parity relates theoretically to existing fairness definitions for regression or structured prediction; discuss mathematical properties.*

Response. We agree this strengthens the contribution. Section 3.2 of the camera-ready paper now situates LP explicitly:

- Relative to **demographic parity / equalized odds**, LP replaces a binary detection outcome with a continuous quality score (IoU), evaluated only on the detected set—analogous to equalized opportunity applied to localization quality.

- Relative to **fair regression**, LP instantiates bounded disparity in expected loss with $\ell = 1 - \text{IoU}$ on matched boxes.
- Relative to **AP-based audits**, LP isolates the localization component that AP conflates with classification confidence and ranking.

We also state clearly that LP is a group-level criterion (not individual fairness or calibration), chosen because the safety harm of interest is group-structured. Orthogonality to detection parity is formalized in Eq. (1) and the surrounding text.

R1.2 — Confounders and statistical methodology

Comment. *Specify the confounders considered and the statistical methodology used.*

Response. Section 4.4 now states this explicitly:

- **Primary confounder:** object scale (ground-truth box height), because female and child pedestrians tend to occupy smaller image regions.
- **Secondary confounder:** occlusion, used where annotations permit.
- **Method:** ICON² stratified binning into short / medium / tall height terciles within each dataset; per-group mean IoU recomputed inside each stratum.
- **Test:** two-sided Mann–Whitney U (non-parametric; IoU distributions are non-Gaussian), significance at $p < 0.05$.
- **Result:** gender gap survives in 35/36 stratified cells; age gap in 24/30.

R1.3 — Practical safety implications of 1–3 pp IoU gaps

Comment. *Discuss how localization disparities of approximately 1–3 percentage points propagate into downstream perception or planning errors.*

Response. New Section 3.3 provides a first-order geometric argument: under pinhole projection, relative depth error scales with relative box-height error. For a pedestrian at 30 m with an 80 px box height, a ~ 2 pp mean IoU deficit (roughly 2–4 px systematic height / center error) corresponds to approximately 0.7–1.5 m of depth bias—enough to alter time-to-collision estimates at urban speeds. Because trajectory and collision modules are typically nonlinear in box geometry, group-level mean gaps of this magnitude remain safety-relevant even when every detection clears the 0.5 IoU threshold. We openly note that a full closed-loop propagation study is future work, now listed in the Conclusion.

R1.4 — Generalization beyond gender/age and autonomous driving

Comment. *Unclear whether the framework generalizes to other protected attributes or domains beyond autonomous driving.*

Response. LP is domain-agnostic: any detector that outputs boxes and any protected attribute with instance-level labels can be audited under Eq. (1). Empirically, this paper studies gender and age in driving scenes because those labels are available at scale. The Ethics and Conclusion sections now state clearly that skin tone, intersectional groups, disability, and non-driving domains (surveillance, robotics) are planned extensions, and explain why skin tone was excluded from the present audit (insufficient labeled sample / coarse categories in our pilot).

We thank Reviewer 1 for the positive overall assessment and for highlighting the value of transparently reporting the negative generalization result for DWBBR; that discussion is retained and sharpened in Section 5.2.

Response to Reviewer 2 (Score: 1 — Weak Accept)

R2.1 — Datasets must be specified

Comment. *No dataset is mentioned; data source, size, and composition must be specified.*

Response. Addressed throughout. The abstract and Section 4.1 now name all four datasets with approximate sizes and composition:

- **BDD100K** — US dashcam; ~100,000 pedestrian boxes; binary gender labels.
- **CityPersons** — ~19,000 boxes / 2,975 images.
- **EuroCity-Day** and **EuroCity-Night** — subsets of EuroCity Persons (~238,200 boxes / 47,300 images / 31 cities).

Phase 2–3 training uses a 1,693-image BDD100K subset (423-image validation).

R2.2 — Detector families must be named

Comment. *Detector families are not named in the abstract.*

Response. Addressed. The abstract and Section 4.2 now name:

- **YOLOX** (single-stage, anchor-free),
- **RetinaNet** (single-stage, anchor-based),
- **Cascade R-CNN** (two-stage),

with **YOLOv8n** as the trainable backbone for the DWBBR intervention.

R2.3 — Demographic axes / race–skin tone

Comment. *Unclear whether race/skin tone was one of the two demographic axes; axes should be stated explicitly.*

Response. The two axes are **gender** (female vs. male) and **age** (child vs. adult). Race / skin tone is **not** one of them. This is now stated explicitly in the abstract, Introduction (contribution 2), Section 4.3, and Ethics. We briefly explain the exclusion (pilot sample too small / categories too coarse) and list skin-tone LP as future work.

R2.4 — Conference template

Comment. *The PanAfriCon conference template has not been used; please reformat to the required Springer LaTeX template.*

Response. Completed. The camera-ready manuscript is prepared with the official Springer LNCS class (`llncs.cls`, bibliography style `splncs04`). Source and compiled PDF are included in the submission zip.

Closing

We believe the revised manuscript fully addresses the reviewers’ concerns while preserving the core contributions that both reviews found valuable: a new fairness criterion, a broad confounder-controlled audit, and transparent reporting of a mitigation that succeeds in training but fails to generalize. We are grateful for the opportunity to improve the paper and look forward to presenting this work at PanAfriCon AI 2026.

Best regards,

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